TABLE 1
SEEBECK COEFFICIENT OF SOME COMMON THERMOELECTRIC MATERIAL IN μV/K

Bismuth	Constantan	Germanium	Silicon	Tellurium	Selenium
-72	-35	330	440	500	900

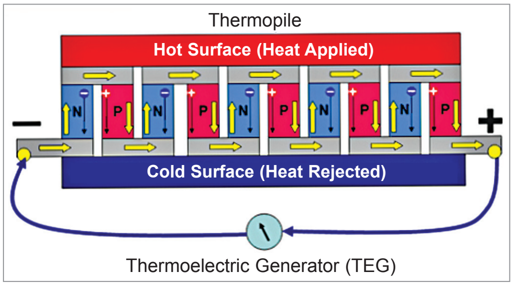

Fig. 5: Series-connected thermoelectric generator

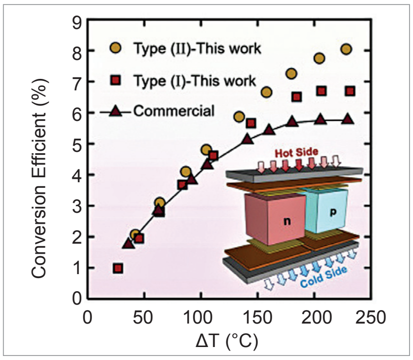

Fig. 6: Conversion efficiency of improved BiTe material

a voltage potential that is directly proportional to the temperature difference across the semiconductor.

The power generated in a TEG is single-phase DC that equals I^2R_L, where I is the current and R_L is the load resistance. The output voltage and output power are increased either by increasing the temperature difference between the hot and cold ends or by connecting several TEGs in series, as shown in Fig. 5.

The current flows as long as heat is applied to the hot junction. The process is reversible. If the hot and cold junctions are interchanged, the valence electrons flow in opposite direction and direction of the current changes. The thermoelectric effect allows converting waste heat into electric power.

By combining thermoelectric and PV effects, higher solar electricity conversion efficiency is possible. PV absorbs about 58% of solar energy between 200nm and 800nm wavelengths. The rest of the solar energy from 800nm to 2500nm cannot be converted to electricity by PV. But this spectrum of solar radiation can generate electricity through thermoelectric effect by heating TEG.

Thermoelectric figure-of-merit

The performance of thermoelectric materials is defined by unitless figure-of-merit as given below:

$ZT = \sigma S^2 T/k$

where ZT is the thermoelectric figure-of-merit while σ, S, k, and T are electrical conductivity, the Seebeck coefficient, the thermal conductivity, and the absolute temperature, respectively.

The Seebeck coefficient of a material is the induced thermoelectric voltage per Kelvin generated in response to a temperature difference across the material, as induced by the Seebeck effect. It is often given in microvolts per Kelvin. The Seebeck coefficient depends on factors like temperature, work functions of the two TE materials, electron densities of the two components, and scattering mechanism with each solid. Performance of a TEG is determined by the Seebeck coefficient of the pair of materials forming the TEG.

Table 1 gives the Seebeck coefficient of some standard TE material in microvolt/K at 0°C. An ideal TE material would have a Seebeck coefficient of more than 200μV/K for best voltage generating ability.

Peltier effect

In a circuit, when DC current flows through two dissimilar material, say copper and bismuth, the junction where the current passes from copper to bismuth would be hot and the junction where current passes from bismuth to copper would be cold. This effect, known as Peltier effect, is used to build devices like Peltier heater, solid-state refrigerator, and heat pump.

A good thermoelectric material should have following qualities:

1. Its Seebeck coefficient should be as high as possible. It is important to maximise energy conversion. The open circuit voltage generated by a TEG is proportional to the Seebeck coefficient and temperature difference across the TEG. Hence high Seebeck coefficient leads to a high voltage.

2. To minimise thermal loss through the thermoelectric material and to have large temperature difference across the TEG, thermal conductivity of the TE material should be as low as possible.

3. Its electrical conductivity should be as high as possible for reducing internal Joule heating losses of the thermoelectric elements.

A TEG's efficiency depends very much on the operating temperature difference between the junctions. The bigger the temperature difference, the more efficient the TEG.

There are three main types of thermoelectric materials used in thermoelectric generators:

1. Bismuth telluride (Bi_2Te_3) alloy. It is a semiconductor that has high electrical conductivity but is not good at transferring heat. The best working temperature of this class of material is below 450°C. Bismuth telluride materials with high figure of merit (ZT) and TEG modules, as shown in Fig. 6, have high conversion efficiency of more than 8% over temperatures of 25°C to 250°C and are widely utilised in energy generation and refrigeration.